NATIONAL AGRICULTURAL RESEARCH PROJECT

SURVEY OF PLANT DISEASES IN KENYA

A Report by Dr. L. E. Claflin

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SURVEY OF PLANT DISEASES IN KENYA WITH SPECIAL ATTENTION TO THE INCIDENCE OF MAIZE STREAK

BY

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MAIZE STREAK DISEASE

Maize streak disease poses a very grave threat to production of maize in Kenya. In fact, vast hectares around the National Research Center at Muguga including the Wangige, Banana (town), Kanunga, and Kiambu areas will not be harvested due to maize streak. Yields will be greatly reduced in other fields that were infected later in the growing season. This disease was present in nearly 20% of the plants examined in the lowlands (Mtwapa), 15-20% incidence in the Embu region, and relatively low (5 - 10%) in the arid region (Katumani).

Maize streak was most prevalent in the higher altitudes as the Muguga region and the vicinity around the tea growing area of Kamweti (Kirinyaga District) will be seriously affected. In the past, the higher altitudes were minimally affected with this disease. Perhaps this change is attributable to 1. a different genus or species of vector that is more adaptable to these areas; 2. a different strain of the virus which is more aggressive; 3. the widespread plantings of napier grass which is used for forage and, also may be a host for the vectors and virus during the year when maize is not growing. In addition, the germ plasm utilized by farmers possesses limited or no tolerance to either the vectors and/or virus. Maize is commonly grown in some areas throughout the year which ensures an everpresent population of vectors and virus.
Maize streak is a difficult disease to study due to the complexity of the interaction between virus and vector. As maize streak occurs throughout Africa, it is imperative that KARI personnel plan and implement cooperative projects with scientists from other countries and the International Centers. Initially, KARI maize breeders, entomologists, and plant pathologists should establish contacts and visit ongoing maize streak research at one or all of the following: CIMMYT in Harare, Zimbabwe; IITA in Ibaden, Nigeria; and, ICIPE in Lusaka, Zambia. Establishment of cooperative projects would avoid duplication of research efforts and allow a free exchange of ideas and germplasm.

The only practical solution is to incorporate tolerant germ plasm into Kenyan composites and hybrids. Germplasm reported to be immune or tolerant to maize streak has been reported from Burundi, Reunion, IITA (Ibadan, Nigeria),and CIMMYT. A number of accessions from Burundi are growing at the Muguga NARC. These accessions will be evaluated at or near physiological maturity. Preliminary observations between maize entries in the National Performance Tests and the Burundi accessions show differences as the latter entries are more tolerant to maize streak.

While surveying farmers fields in the highlands near Muguga, maize plants of a local landrace and some plants of Katumani Composite expressed tolerance to maize streak. These plants were systemically infected with the virus, yet were significantly taller and had well developed ears in comparison to adjacent infected plants. The plants were marked and will be harvested at physiological maturity. This germplasm will subsequently be tested for tolerance to maize streak under glasshouse conditions.

Farmers have limited or no knowledge about maize streak. Some have attributed the cause to a conspiracy by Western Kenya maize growers who have, by some fashion, delivered a plague to destroy the maize whereby Eastern Kenya farmers will have to purchase maize from those in the west. Others believe the cause is due to faulty fertilizer. It would be most advantageous if the Extension Service would disseminate the facts concerning maize streak by all available means of communication.

Screening germplasm for tolerance to maize streak requires facilities that are not presently available at Muguga. To provide credence to the identification of tolerant accessions, mandates that the testing be conducted under screenhouse and/or glasshouse conditions due to the complex interaction of the leafhopper vector and virus. This would ensure that the plants were exposed to the viruliferous leafhoppers and, more importantly, the plants would all be exposed at the same stage of growth. Once tolerant accessions are identified, they would be released to other NARC centers for testing. A glasshouse presently in place at Muguga could be renovated and enlarged to accomodate the screening assay. Excellent glasshouse and screenhouse structures have been constructed recently at Thika that are very functional and were erected at a modest expenditure. In addition, the ICIPE facilities (Mbita Point) are superb and were designed primarily for insect research. It would be advantageous for KARI to consider the design of these structures when funds become available.

It is imperative that maize germplasm developed at the various centres breeding programs be evaluated for disease tolerance at all centres locations prior to release. This will partially alleviate concerns that races or strains of pathogens exist over the vast expanses of maize production in Kenya. Strains and/or races of maize streak, common rust, and northern corn leaf blight have been reported from numerous countries.

Maize inbreds and cultivars will need to be imported from International centres and other countries to accelerate the release of maize streak resistant germplasm for Kenya farmers. The Quarantine center at Muguga must expedite release of this germplasm to the breeding programs post haste. The glasshouse facilities at Muguga are too limited for an extensive "grow-out" of imported seeds. Establishment of an isolated plot within a national forest that is at least several kilometres from established maize production will hasten the quarantine protocol. The present procedure is too restrictive as maize seed obtained from IITA in October, 1988 has not been released. Enlargement of present facilities or construction of additional

glasshouses and/or screenhouses at Muguga will alleviate the problem. In any event, the seed should not be quarantined over six months.

Maize streak is not only a threat to Kenya maize production in the future, it is causing a disaster at the present. The potential of this disease far exceeds the southern corn leaf epiphytotic that devastated the maize crop in North America in 1970.

Malfunctioning Equipment:

Laboratory equipment is a vital and expensive component in any scientific research program. Scientific equipment is minimal in most KARI laboratories, however, in numerous instances, the equipment on hand was not being utilized due to various malfunctions. It would be cost effective to have trained personnel and an inventory of replacement items to fix the equipment at a central location, e.g. Muguga. For example, two of the ultracentrifuges at Muguga are not being used because of faulty vacuum pumps. I was told that technicians and the replacement items to fix the centrifuges must be brought in from Vienna, Austria. Perhaps an engineer or a graduate from polytechnic school could be trained at the location of the manufacturer for several months. The same problem is experienced with microscopes as they have to be sent out of Kenya for repair and the delay may be up to one year or more. It would also simplify the situation if the equipment purchased was from one manufacturer. In the case of microscopes, purchase those made by Olympus or Leitz, etc. Seemingly, other governmental agencies such as the Kenya Medical Research Institute, Universities, and perhaps private companies could also utilize these services and be charged accordingly.

Equipment Sharing among Institutes:

It is prohibitive to purchase expensive equipment such as scanning and transmission electron microscopes, ultracentifuges and numerous other items for every research center. Perhaps if the equipment is in place at one Institute, they could charge a minimal fee for other Institutes to utilize the equipment. For instance, the Kenya Medical Research Institute has two transmission electron microscopes that could be used by the Plant Virologists for researching virus diseases of maize. This equipment sharing will become more important as scientists become involved in biotechnology.

Careers of Governmental Retirement age For Scientists;

Well trained scientists that have earned Ph.Ds may retire at age 45 and mandatory retirement at 55. This appears to pose a serious problem as many scientists are in their 30s when the Ph.D is granted. As a result, the scientist may have only ten years of service if they elect to take an early retirement. This would tend to result in a constant turnover of experienced personnel that are so vitally needed in a research institute. Perhaps nothing can be done except that monetary rewards be considered for those who remain productive and stay in leadership and/or scientific roles.

Successful Seed Production Programs in Kenya

A large number of potentially devastating diseases are seed-borne. This is particularly true for important Kenyan staples such as dry beans and cowpeas. In examining these commodities for disease incidence, I was unable to detect any viral or bacterial diseases that are primarily disseminated via the seed. Yet, in other African countries, the crops have been devastated by seed-borne diseases. In Kenya, strict adherance to the Zero or minimal tolerance levels by various research centers such as the National Horticultural Research Center at Thika and the Seed Quality Research Center at Lanet are responsible for this success. These seed certification programs have contributed far more to the economy by reducing yield losses than the costs incurred in maintaining the programs. The Kenya Seed Company should also share in the success for their role in distributing seeds that possess vigor, have varietal purity and maintain rigid tolerances for seed-borne diseases.

One seed certification program that may face potential problems in the future pertains to the Irish Potato. Bacterial ring rot is a very serious disease and is disseminated primarily through the vegetatively propagated tubers. In the Embu area, numerous farmers complained of bacterial ring rot. Not only is this disease a problem in lowering yields, it also contributes to early decay of tubers in storage. The Problem in Embu is compounded by the extensive production of potatoes and that "volunteer" potatoes are present throughout the year which serves as a reservoir for the bacteria. Bacterial ring rot nearly destroyed the potato industry in North America during the 1940s. Adherance to a very strict (zero tolerance) certification program was soley responsible for the continued viability of potato production in North America.

Sanitation is also an important aspect in controlling bacterial ring rot. Equipment used in planting and harvesting the crop and storage facilities must be thoroughly disinfested with chemicals such as formaldehyde and sodium hypochlorite. Volunteer potatoes should be rogued on a regular basis. In addition, after bacterial ring rot is detected, potatoes should not be planted in that location for two years. An additional precaution should be addressed; bacterial ring rot may be very difficult to diagnose at Tigoni due to the altitude and climate and it may be necessary to conduct the testing at lower altitudes with the additional stresses.

Disease Performance Test

The Maize National Performance Test (NPT) is planted at numerous locations throughout Kenya for the purpose of ascertaining composites and hybrids that exhibit desirable agronomic characteristics for a particular locale. In conjunction with the NPT, it would be desirable to establish a Disease Performance Test. This test would consist of planting known susceptible and resistant germ plasm for the following diseases: Maize streak, northern corn leaf blight, head smut, common rust, and southern rust. The Disease Performance Test would be invaluable in determining the important diseases within a particular local, for training personnel in identifying maize disease by symptom

expression, for monitoring disease incidence, and the test would provide an early warning system if new races or strains of the pathogen are prevalent.

Kenya Leaf Blotch

Kenya leaf blotch is a disease of unknown etiology that was observed throughout the country. The incidence was more pronounced in several areas including Mtwapa, Muguga, and Embu. In addition, certain hybrids and lines appeared to be more susceptible. The symptoms are characterised by small (2 - 3 mm), yellow, translucent necrotic areas that resembles a Zonate appearance. Often, the necrotic areas were water-soaked around the margin. The lesions coalesced which resulted in a relatively large section of the leaf affected.

Sections of leaf tissue were placed in a humidity chamber overnight and then observed for fungal and/or bacterial pathogens. Bacteria were observed streaming from the vascular tissue and, consequently, may be the likely candidate as the causal agent. In North America, a disease known as Holcus Spot, is caused by the bacterium, Pseudomonas syringae Pv.syringae. Holcus spot symptoms closely resemble those observed in Kenya maize.

Samples were collected and efforts will be devoted to identify the causal agent in my laboratory.

RECOMMENDATIONS AND SUGGESTIONS

1. Prohibit planting of maize other than during the normal planting dates for the long rain and short rain seasons. At the present, maize is planted continuously throughout the year. I have noticed that maize streak is more prevalent in those regions where maize is grown throughout the year. In most vector-pathogen systems and in the life cycles of mammalian vector-parasites, the control strategies include interrupting the weakest link in these complex interactions. Allowing one month or more between harvest and planting the next maize crop would disrupt either the leafhopper life cycle or, most likely, reduce or eliminate the availability of maize streak virus inoculum. This precludes the possibility of other host plants in these regions such as other cereal crops or grasses.

   This control strategy is not unique to agriculture as an extensive program was implemented in North America to eliminate barberry plants which are the alternate hosts for stem rust of wheat. Another example involves the elimination of Johnson grass which is an alternative host of Maize Dwarf Mosaic Virus (Strain A) during the non-growing season of maize in North America. Prohibiting the planting of maize except during the bimodal seasons should not impose a hardship on the majority of Kenya farmers or seriously affect the availability of maize. Seemingly, maize produced during the off-season is mostly for speciality purposes such as roasted maize. If adopted, penalties should be decided by KARI and the Ministry of Agriculture.

2. It is imperative that KARI scientists plan and implement cooperative projects within the KARI framework and also with scientists from other countries and International Centres. In particular, nearby centers such as CIMMYT in Nairobi, and ICIPE (Mbita Point) should be included in maize streak control and breeding strategies. It would be cost effective for KARI to send a breeder, plant pathologist, and an entomologist to one or all of the following programs with maize streak projects: CIMMYT in Harare, Zimbabwe; IITA in Ibaden, Nigeria, and ICIPE in Lusaka, Zambia. These cooperative projects would allow for exchange of ideas, research results, and germplasm.

3. Germplasm will need to be imported. For the immediate future, germplasm from IITA (mid-altitude lines from Cameroon) and CIMMYT, Harare, Zimbabwe would be likely choices for Kenya. They will need to be tested for maize streak resistance in Kenya as there is the distinct possibility that different strains, genera, or species of leafhopper vectors or different strains of maize streak exist. In addition, several other lines or cultivars presently grown in Kenya are viable choices. These include Titikaca (from Peru), Kikuyu landrace that was identified as tolerant in Kiambu, and the composite, Katumani. Facilities are limited for screening these materials at the present. Perhaps arrangements could worked out with ICIPE personnel at Mbita Point and/or attempt to locate space in screenhouses at other KARI stations, e.g. NARC at Thika.

4. Germplasm from IITA, Reunion and Burundi are unacceptable at present. The IITA and Reunion lines are not adapted for mid-altitudes. The Burundi material is a segregating population, thus it will require 4-5 years for selecting desirable lines.

5. Biotechnology could play a pivotal role in researching maize streak virus. The following ideas are not "stop gap" measures but are of long term and worthy of the expenditures:

   a) Monoclonal antibodies are most specific and would be an important aspect in researching the virus-vector interaction, ascertaining strains of maize streak, and to determine other potential hosts such as napier grass. Monoclonals are time-consuming to develop and expensive, however, once specific antiserum is identified from the hybridoma, the quantity of antiserum is limitless.

   b) Tolerance in maize germplasm is essentially based on symptom development, plant height and ear formation in comparison to plants of susceptible genotypes. Therefore, tolerance may or may not be indicative of the number of virus particles in a measured quantity of expressed plant sap or virus replication within the maize plant. This would be determined by extraction and purification of the virus and then performing a polyacrylamicide gel electrophoresis (PAGE) experiment to determine the amount of protein present in the corn tissue. An increase in protein would indicate a greater number of virus particles.

      Development of cdna (complementary) probes might be more cost effective than monoclonal antibodies and, yet, be as equally effective in experimentation.

      Polyclonal antisera is available and could be incorporated into recently developed diagnostic assays such as Enzyme linked immunosorbvent assays (ELISA) or dot-immunobinding assays.

   c) Reombinant DNA strategies are being tested in maize in North America. One particular experiment involves inserting a Bacillus thuringensis (Bt) gene into a plasmid of E. Coli. This plasmid was then inserted into the bacterium, Clavibacter xyli subsp. cynodontis. This bacterium is a pathogen of bermuda grass and will also colonize the vascular bundles in maize without any apparent effect on the plant. The Bt gene produces a toxin that is effective against chewing insects, thusly the purpose of the recombinant DNA experiment is to provide the maize plant with a built-in insecticide against borer. The bacteria is applied as a seed dressing.

      A similar strategy might be attempted in maize by using the coat protein of the maize streak virus. This would assumingly tie up receptor sites for virus attachment in the host plant and thus prevent the intact virus particle to

attach. Binding of the virus particle is an initial step in the infection process.

6. Implementing a Disease Performance Test along with the National Performance Test would be an invaluable aid for determinning prevalent maize diseases in a particular region, training personnel in disease diagnostics, monitoring disease incidence, and provide an early warning system if new roles or strains of the pathogen should arise. The test would consist of planting susceptible and resistant genotypes for a particular disease. My suggestions for selecting the genotypes and the potential reaction to a particular pathogen are as follows:

   Maize streak virus

   H614 - susceptible check
   CIMMYT Pool 431 or Titikaca - Resistant (?)

   Northern corn leaf blight

   GSO7(R)C2 - Susceptible check
   Ecuador 573 - Resistant

   Southern corn leaf blight

   CIMMYT Pool 430 - Susceptible
   SR52, line 24 - Resistant

   Head Smut

   H614 - Susceptible check
   SR52 - Inbreds 3 or 5 - Resistant

   Common Rust

   Makweni - susceptible
   H622 - resistant

   Kenyan leaf blotch

   SR52, line 17 - susceptible
   H512 - resistant

   Southern rust:

   No recommendations as the disease was observed in only one location.

   Caution: These genotypes are not adaptable to all locations and these recommendations are based on only one observation at different locations in Kenya.

7. Investigate other potential viral problems in maize such as maize mosaic, sugarcane mosaic and maize dwarf mosaic. These viral

pathogens may be interacting in a synergistic fashion with maize streak.

8. Kenyan leaf blotch, a disease observed throughout Kenya, is of unknown etiology. The causal agent is suspected of being a bacterium. Numerous samples were collected and efforts will be devoted to determine the pathogen. Other possibilities include mycoplasma, a spiroplasm, viruses or, perhaps a genetic defect.

9. Laboratory facilities and/or scientific equipment for plant pathology research is almost nonexistant. Hopefully, with support from USAID, MIAC, and the World Bank, this situation will be resolved. At the present, it is nearly impossible for the scientists to evaluate germplasm and to prioritize their research as they have neither adequate facilities or equipment to accomplish the tasks.

10. An additional plant pathologist is needed to research diseases of cereal crops at Katumani. The incumbent plant pathologist is presently studying for a M.S. degree in Holland and will be be assigned to legume crops upon his return.

11. Everyone must cooperate in solving the maize streak epiphytotic in Kenya. Whoever releases a maize hybrid or cultivar that is of desirable agronomic traits with resistance to maize streak disease will likely control the African maize seed market.

12. With a risk of failing to mention a particular name or location, I wish to thank everyone associated with KARI, MIAC and USAID for your cooperation and hospitality during my seven weeks in Kenya. It has been a delightful time.
Asante Sana!

SYNOPSIS OF ACTIVITIES AND

RESULTS OF PLANT DISEASE SURVEY

July 14 Departed Manhattan, Kansas, 11:00 am enroute to Nairobi.

July 15 Arrived at Kenyatta Airport at 9:00 p.m. Was met by Dr. Hires and transported to Hilton Hotel.

July 16 Rested at Hotel.

July 17 Met Dr. David Daugherty, Chief of Party, MIAC and transported to Kenya House. Introduced to Dr. C. G. Ndiritu, Director, KARI and discussed my role as a consultant to KARI and MIAC.

Discussed various research projects with Dr. Lee Kolmer and Dr. I. M. Gathuma (Faculty of Vet. Med., Univ. of Nairobi) who were finishing consulting assignments to MIAC.

Discussed visits to KARI centers with Dr. Larry Empig and Dr. Daugherty for the next seven weeks.

Visited very briefly with Dr. Vartan Guiragossian, sorghum breeder for East Africa, ICRISAT (located in Nairobi) concerning the covered kernel smut of grain sorghum research project at Muguga. Plans were made to meet at Muguga to discuss inoculation techniques.

July 18 Travelled to Katumani with Dr. Larry Empig and was introduced to Mr. Kusewa, Director; Dr. Karie Njoroje and Mr. Cesar Grathuri, Maize breeders; and Mr. Lawrence M'Ragwa, millet breeder. Very limited time was spent at Katumani as an extended visit is scheduled for July 24-26.

A visit was made to the National Range Research Center at Kiboko which is at an altitude of 950 m and receives 640 mm (25.2 in) of rainfall. Maize grown at this station is primarily for seed. The ears were well developed although several exhibited yellow kernels which indicated cross-pollination from adjacent fields. To ensure varietal purity, the seed fields should be isolated from other maize varieties.

Common rust and southern corn leaf blight were observed and were of minor importance. Kiboko appears to be an ideal location for evaluating maize germplasm for stress (drouth) tolerance.

ICRISAT has extensive plantings of sorghum and pearl millet at Kiboko. Sorghum downy mildew and rust were observed on sorghum. Rust and ergot (sugary disease) were observed on

millet.

Departed Kiboko around 11:00 a.m. and arrived in Mombasa around 5:30 p.m.

July 19 Arrived at the Mtwapa station around 8:30 a.m. and was introduced to Mr. Aziz, Station Director; Mr. Winston Shibatsi, maize breeder; Mr. Kiarie Mwangi, Entomologist; and Mr. Symon Mwangi, Plant Pathologist. Several hours were spent evaluating existing laboratory equipment and future needs with Mr. Mwangi. Time was also spent assessing existing equipment and repairs of nonfunctioning items with Dr. Bill Hires.

Mr. Shibatsi's plots were surveyed for disease incidence. Common rust was the most prevalent with lesser amounts of northern corn leaf blight, maize streak and curvularia leaf spot. An unidentified leaf disease characterized by small, yellow, translucent, neurotic areas which is similar to Holcus spot in North America was of high incidence. Until the disease is positively identified in my laboratory, it will be referred to as Kenyan leaf blotch.

July 20 Dr. Empig, Mr. S. Mwangi, Mr. Shibatsi, and myself inspected the maize and grain sorghum accessions of the Kenya Seed Company at their Baobab farm. Mr. Albert Sumba hosted us on the tour. The crops were relatively free of diseases. Curvularia leaf spot and Kenyan leaf blotch predominated. Minor incidences of northern corn leaf blight and common rust were observed in maize. No identifiable diseases were observed on sorghum. The crops were excellent and were being grown under high fertility.

Mr. Shibatsi's plots at Kambe and Rebe were examined for diseases and agronomic characteristics. Common rust was prevalent at Rebe with lesser incidences of Kenyan leaf blotch and northern corn leaf blight. Only Kenyan leaf blotch was identified in the Kambe plot.

July 21 Dr. Empig, Mr. Mwangi, Mr. Shibatsi, and myself toured the Msabaha station near Malindi. The soils are very sandy with limited organic matter. Mr. Waweru Gitonga, Director, discussed the objectives of the Msabaha station and provided a tour. Due to untimely rainfall and sandy soils, the maize plants were noticeably stunted and will yield below average. Maize streak and Kenyan leaf blotch were the only diseases observed. Mr. Gitonga will soon be leaving to pursue a M.S. degree in entomology at a university in England (NRC Scholarship).

July 22 Toured experimental plots of cassava, cocoa, cowpeas, sorghum, pigeon peas, sugarcane, grapes, passion fruit, and Irish potatoes with Mr. S. Mwangi. The more important diseases are as follows:

Irish Potato: Potato virus X, blackleg (bacteria).

Cassava: African cassava mosaic virus, 100% incidence - very severe.

Sugarcane: Virus disease - probably one of the sugarcane mosaic viruses; unidentified malady characterized by stunted and chlorotic plants with a striping pattern; possibly due to fertility problems or gumming disease (bacteria).

Cowpeas: No disease observed which is attributable to the excellent seed certification programs in Kenya.

Mr. S. Mwangi and I discussed the maize screening program at length. Approximately 2,500 sq. meters at Mtwapa will be developed for irrigation of which 500 sq. meters will be devoted for screening germplasm for generic variability to southern rust and southern corn leaf blight. Mr. Mwangi will be responsible for planting susceptible spreader rows around the plot and two rows between every eight rows of germplasm being evaluated. The spreader rows will be planted three weeks prior to planting the germplasm evaluation which will ensure adequate inoculum. In addition, Mr Mwangi will plant several susceptible genotypes throughout the year for maintaining inoculum during the dry season. The Mtwapa station is a viable location for testing germplasm against maize streak. The vector (leafhopper) is certain to proliferate in the coastal area and maize streak has other hosts besides maize. These include wild grasses, sugarcane, millet, cereal crops, and rice. Mr. Mwangi stated that maize streak is endemic in the lowlands and he has observed upwards to 40% incidence. All maize lines, composites and hybrids should be evaluated at Mtwapa for maize streak as several strains of the virus have been reported.

This station would also be an ideal location for evaluating sorghum germplasm for grain quality (weathering, diseases, etc.) and stalk rots.

July 23 Departed Mombasa at 8:00 a.m. and arrived in Nairobi around 3:00 p.m.

July 24 Rested at Hilton Hotel

July 25 Dr. Empig and I visited briefly with Dr. Daugherty before proceeding to Katumani. Discussions were held with Mr. Kusewa, Dr. Njoroje, Mr. Grathuri, Mr. M'Ragwa, and Mr. Clement Kamau (sorghum breeder). This national dryland farming center is currently without a plant pathologist as Mr. Wilson Songa is pursuing a M.S. degree in Holland and will not return until 1990. After finishing the M.S., he will be researching problems on legumes and commodities other than cereals.

A plant pathologist is needed to work on cereal crop diseases due to the excellent ongoing work in breeding early maturing maize hybrids for semi-arid areas. Also, this station is contributing greatly to the Kenyan farm economy by evaluating sorghum accessions for cold hardiness and drouth tolerance. This necessitates the need for a plant pathologist to work exclusively with maize, sorghum and pearl millet.

Northern corn leaf blight was of high incidence in several of the plots at Katumani and appreciable yield losses can be expected. This fungal pathogen was especially aggressive as lesions extended up to 30 cm and were 1-3 cm in width. Most of the leaves including the flag leaf exhibited lesions.

Common rust was particularly severe on several accession, notably Makweni. Here, this composite would be a good choice for a susceptible check or for use in spreader rows. The Katumani station would be a good location for screening maize germ plasm for common rust and northern corn leaf blight.

July 26 Scheduled to return to Katumani but the visit was cancelled as the scientists were attending a meeting in Nairobi.

A meeting was scheduled with Mr. David Court, the Rockefeller Foundation, in Nairobi. Discussions centered around Rockefeller projects. In East Africa, the only ongoing project involves bananas in Uganda. Their emphases is presently on biotechnology. It is uncertain if Rockefeller will fund additional East Africa projects in the immediate future.

July 27 The scheduled meeting in Katumani was cancelled.

A tour of the new facilities of the Kenya Medical Research Institute was conducted by Dr. Davy Koech, Director. KEMRI is housed in a new complex that is fully equipped with state of the art scientific laboratory equipment. The purpose of this visit was to inquire about the prospects of KARI scientists using scanning and transmission electron microscopes. Dr. Koech was receptive for allowing use of the transmission E.M. for plant virology research on a "need be" basis when KEMRI wasn't using it. He suggested contacting ICIPE in Duduville; KEMRI and the Primate Research Lab. in Nairobi, and ICRAF in Gigiri about using the scanning E.M. Perhaps other sophisticated and expensive equipment at KEMRI could be utilized by KARI scientists in research projects.

July 28 Departed Nairobi at 8:00 with Dr. Mick O'Neil; arrived in Embu at 10:30 a.m. Discussed objectives of the Embu NARC with Mr. Gachanja, Director and Mr. Linus Muriithi, plant pathologist. Maize is the major thrust of the Embu station

with 20% devoted to early maturing hybrids and 80% for development of materials suitable for arid and semi-arid areas. Maize streak receives top priority for disease research with lesser emphasis on head smut, common rust and northern corn leaf blight.

The maize National Performance Test at Embu showed the following disease incidence (% of plants infected):

Northern corn leaf blight	90-95
Common rust	30-40
Kenyan leaf spot	25-30
Curvularia leaf blotch	10-15
Maize streak	15-20
Maize streak and maize mosaic (combination)	5

July 29 Surveyed the mid altitude region near Kamweti (Kirinyaga District) with Mr. Muriithi, Mr. Z. K. Muthamia, maize breeder, and Drs. O'Neil and Empig.

Farmer's field:

Maize streak	30%
Northern corn leaf blight	100%
Kenyan leaf blotch	20%

Farmer's Training Program, Ministry of Agriculture, Kamweti, Miss Nancy Mugo, Vice-Principal was the host. This location (um-1 zone) is at 5,800 ft. and receives an average of 2,250 mm (88 inches) of rainfall with temperature ranging from 13 to 18 c.

Maize streak	40%
Kenyan leaf blotch	70-80%
Zonate leaf spot	Trace
Pokka boeng	1-2%
Northern corn leaf	100%
Curvularia leaf spot	20-25%
Maize bushy stunt (?)	1 plant
Sorghum downy mildew (?) in entry No. 27. Important to confirm	

Kutus (K.T.I.)

Northern corn leaf blight	100%
Kenyan leaf blotch	30-40%
Pokka boeng	1-2%
Common rust	5%
Curvularia leaf spot	20-25%

July 29 Discussions were held with Mr. Muriithi concerning the following:

1. Prioritizing laboratory equipment for the Embu station;

2. Protocol and methodologies involved in screening maize germplasm for northern corn leaf blight, maize streak, and Kenyan leaf blotch;

3. Identification and diagnostic techniques that can be utilized for ascertaining maize diseases in the Embu region;

4. The Farmers' Training Program at Kamweti is an optimum location for screening germplasm to maize streak and northern corn leaf blight;

5. Prospective universities in the U.S. that Mr, Muriithi might consider for a Ph.D. program and the major emphasis for a thesis project.

July 30 Rested at hotel.

July 31 Surveyed maize fields at Mweiga town (near Nyeri).

Farmer's field No. 1

Maize streak	60-65%
Head smut	2-5%
Rust	trace
Northern corn leaf blight	trace

Farmers' field No. 2

Head smut	5-10%
Maize streak	75-80%
Common rust	trace
Northern corn leaf blight	trace

Farmers' field No. 3

Maize streak	100% (Hybrid 614)
Head smut	.0
Northern corn leaf blight	trace

Farmers' field No. 4

Maize streak	80-90%
Head smut	0
Northern corn leaf blight	trace

This particular location has extensive production of Irish potatoes. Farmers were complaining about bacterial ring rot which is a tuber-borne disease. An extensive program involving sanitation and planting disease-free seed are the only alternatives for control.

Aug 1 Mr. Rubui, maize breeder and Mr. Muriithi and myself surveyed in the vicinity of Muranga which is a very arid region (500 mm rainfall). Maize, sorghum, and pearl millet plantings were severely drouth stressed with negligible

plant diseases. Most of the crops had been harvested and yields were very low. The farmers can expect to harvest only 2-5 bags/hectare (1 bag=90 kg) of maize. Sorghum and pearl millet are usually grown during the short rainfall period (September-November).

Charcoal rot (Macrophomina phaseolina) was observed in most cowpea plants. This disease is most likely a serious constraint to cowpea yields in the Muranga area.

At the request of the Ministry of Agriculture, Agricultural Society of Kenya (ASK), we were requested to survey the farmer demonstration plots at the show grounds for disease incidence. Our readings are as follows:

French beans:	high percentage of plants with virus; false rust; rust.
Tomato:	Virus disease in every plant - probably big bud.
Kale:	Virus disease - about 50% of the plants affected.
Maize:	none
Sorghum:	none
Bananas:	none
Cassava:	none

Aug 2 Examined the maize plant introductions, inbreds and lines at Embu with Mr. Rubui and Dr. Empig and readings as follows:

(Ratings: 1 = no disease; 6 = Severe disease)

CIMMYT Pool 431 (mid altitude, flint-type, composite)

very clean	
Kenyan leaf blotch	trace

SA Photo Syn II (early maturity)

Northern corn leaf blight	4.5
Kenyan leaf blotch	trace
Curvularia	trace

SA Photo III

Northern corn leaf blight	4.5
Kenyan leaf blotch	trace
Curvularia	trace

K305 - Va 26

Northern corn leaf blight	3.0
Very short, good leaf angle	

K306 - H96
Northern corn leaf blight 4.0
Very short

Titikaca (Peru)
Northern corn leaf blight 2.5
Good color, no maize streak

GSO7(R) C2
Northern corn leaf blight 4.5

Embu 11
Northern corn leaf blight 4.0

PR 8530 (Poza Rica) x E11
Northern corn leaf blight 4.5
Common rust 3.0

Ecuador 573
Northern corn leaf 1.0
Common rust 1.0

CIMMYT Pool 430
Northern corn leaf blight 2.0
Southern corn leaf blight 3.5

CIMMYT Pool 20
Northern corn leaf 3.0
Common rust 2.0

Kitale II
Nothern corn leaf blight 2.0

The following are lines derived from S3 SR52

Line 1 : Ht = 3.0; C = 2.0 (a)
Line 2 : Ht = 4.0
Line 3 : Ht = 4.2; C = 2.0
Line 4 : Ht = 3.5
Line 5 : Ht = 2.5; Hm = 1.5
Line 6 : Ht = 3.0
Line 7 : Ht = 4.0; Hm = 2.5
Line 8 : Ht = 1.0; KLB = trace
line 9 : Ht = 2.5
Line 10 : Ht = 2.5
Line 11 : Ht = 3.5; Hm = 3.0
Line 12 : Ht = 2.5; CR = 1.5
Line 13 : Ht = 1.5
Line 14 : Ht = 3.0; CR = trace
Line 15 : Ht = 2.5; CR = 4.0
Line 16 : Ht = 2.5; CR = trace
Line 17 : Ht = 2.0; CR = 2.5; KLB = 4.5
Line 18 : Ht = 3.0; CR = 2.0; Hm = 2.5
Line 19 : Ht = 2.5; CR = trace; C = 2.5
Line 20 : Ht = 3.0
Line 21 : Ht = 3.5; CR = trace

Line 22 : Ht = 3.0
Line 23 : Ht = 2.5; CR = 3.5; Hm = trace
Line 24 : Ht = 4.0; C = 2.5

a = Ht = Northern corn leaf blight;
CR = common rust;
C = Curvularia leaf spot;
Hm = Southern corn leaf blight;
KLB= Kenyan leaf blotch;

Aug 2 Readings were taken of the germ plasm planted in the Embu demonstration trials. Ratings are as follows:

Embu 11: Ht=3.5; CR=3.5; Hm=2.0;
Embu 12: (male parent) Ht=4.5; Hm=2.0; CR=1.5; C=2.0; KLB=3.5
H511: Ht=3.5; CR=trace; Hm=2.5; KLB=2.5
SR52 (single cross female): Ht=3.0; Hm=1.0; CR=trace; KLB=1.0
H512: H511(f) x SR52(m): Ht=2.5; Hm=1.5; CR=1.0; KLB=1.0
Katumani Composite B: Ht=4.5; Hm=trace; KLB=1.0; CR=1.0
H622: Kitale syn II(f) x Ecuador 573(m): Ht=2.5; CR=1.0, KLB=3.0; maize streak=3.5
H613: Ht=1.0; KLB=2.0; CR=trace; C=1.0

Aug 3 Departed Embu and stopped at the National Horticultural Research Laboratory at Thika enroute to Nairobi. Briefly met and visited with Mr. S. K. Njuguna, Director and Mr. Michael Omunyin, plant virologist. Primary purpose of this visit was to examine the new glasshouse and screenhouse facility. Three glasshouses, each approximately 10m wide and 30m long, were constructed for less than one million shillings. These structures would be ideal for screening maize germplasm to maize streak.

We met Dr. Daugherty at the Kenya House and departed for Muguga. We visited with Mr. T. Ochor, Mr. Jackson Njuguna, and Mr. James Theuri and made arrangements for forthcoming maize streak surveys.

Aug 4 Evaluated and took readings of the maize NPT at Muguga with T. Ochor, L. Muriithi, J. Njuguna, and J. Theuri.

A survey for maize streak was taken in the highlands around Muguga. This included the areas around Wangige, Banana, and Kanunga. Most of the fields examined had maize streak in 75% or more of the maize plants. In numerous fields the incidence approached 100%. Many farmers will harvest the maize for forage and not for grain as ear development was nearly nonexistant.

Aug 7 Discussed the Burundi Maize streak germplasm with Dr. Dunston Malithano who is employed by IDRC (Canada) and working as a maize breeder at Muguga. The Burundi materials exhibited far less maize streak than the NPT entries that were planted nearby.

Toured the maize head smut nurseries with J. G. Njuguna. Mr. Njuguna has worked out a successful inoculation procedure by adding 250 mg of teliospores per seed prior to planting. This results in about 60% of the plants becoming infected which is acceptable. Inbreds 3 and 5 of SR52 are resistant and South Carolina 56 is susceptible to head smut.

Mr. Njuguna is also working on an inoculation procedure for long smut of sorghum in cooperation with ICRISAT. The clamydiospores are germinated on PDA media and the spondia is used as inoculum. This work is being conducted in the glasshouses because of the heat units required by sorghum. A low disease incidence usually occurs although accession no. IS8595 has been found to be immune.

The grain sorghum accessions were surveyed for disease incidence with the following diseases recorded:

Bacterial leaf blight (Psudomonas avenae) 50%
Pokka boeng 5%
Leaf spot (unknown causal agent) 80%

We toured the laboratory facilities and discussed plans for building a new screenhouse/glasshouse and/or modifying the existing structure for screening maize germplasm to maize streak.

Aug 8 Visited with Mr. Theuri concerning possible universities in the U.S. for a Ph.D. program.

Visited with Dr. B. W. Ngundo concerning the priorities and objectives of the NARC at Muguga.

A survey for maize streak was conducted in the lower elevations around Muguga. This area is more arid with grazing of animals being the primary income for farmers.

Thiogio:

Maize streak	25-30%
head smut	5%
Northern corn leaf blight	80%
Phyllosticta leaf spot	2-5%

Ngong Hills:

Maize streak	10-15%
Northern corn leaf blight	100%

Aug 9 Returned to the National Horticultural Research Institute at Thika to observe the tomato, bean, and banana accessions at the request of Mr. Michael Omunyin, Plant Pathologist. Bacterial diseases, particularly canker and southern wilt were severe on tomatoes. Farmers are reluctant to plant improved varieties with disease resistance as the old,

indeterminate cultivar, money-maker, is apparently preferred by consumers.

The new glasshouse/screenhouse was re-examined to determine the feasibility of growing maize in soil rather than in pots or benches. Again, these structures are most adequate for screening germplasm for streak resistance and maize can be planted in soil as the height of the structure is sufficient.

The covered kernel and long smut program of sorghum at ICRISAT was discussed with Dr. Vartan Guiragossian. ICRISAT has funded the program at Muguga for several years and are pleased with the results.

Aug 10 Plans were made with Dr. Daugherty concerning the forthcoming trip to Kitale, Kakamega and Kisumu.

Departed Nairobi at 10:00 a.m. and arrived in Kitale at 6:00 p.m.

Aug 11 Met with Mr. Muthoka, director, Drs. Bill and Mary Comptom and Mr. Ochieng, maize breeder.

We were unable to tour nurseries in the afternoon due to rain. Maize breeding methodologies for high altitude areas and disease problems in the Kitale area were discussed.

Aug 12 Dr. Crompton, Mr. Ochieng and myself toured the breeding lines, plant introductions, and national Performance Tests.

Diseases observed included:

Maize streak
northern corn leaf blight
common rust
Pokka boeng
Kenya leaf blotch

A meeting was held in Nairobi for KARI scientists, directors and others involved in agricultural research during August 14 through August 16. As most of the maize professionals would attend the meeting and due to the illness of Mr. Ochor, plant pathologist, it was decided that it would be more beneficial if I traveled to Kisumu and Mbita Point during the week of August 14.

Aug 13 Traveled to Kisumu and stopped at several sugarcane and maize fields enroute.

Aug 14 Made arrangements to tour the Sugarcane Research Station at Kibos. A tour was conducted by Mr. Gerald Ongoma and Mrs. Mary Kariaga, plant pathologists. The primary purpose was to collect plant specimens with gumming disease to supplement our extensive culture collection. Smut was the

principal disease observed. I was unable to meet with Dr. Osoro who was in Nairobi.

Aug 15 Dr. Walter Heid provided a tour of the USAID/DPRA on-farm grain storage project at Kisumu. The thrust of this program is to improve grain storage facilities for the small farmer through the use of the cribs, application of insecticides, and use of cultural techniques for grain drying. These simple but useful facilities often return a profit to the farmer after one year by preventing storage losses. They have purchased over 600 bicycles for use by agricultural offices in various districts. It is estimated that the efficiency in which these officers can visit farmers by use of a bicycle rather than walking is around 200%.

Mr. Francis Kiriro, agronomist and Deputy Director of the Regional Research Centre at Kakamega was host for a tour of this facility. The principal diseases observed include the following:

Maize:
- common rust
- northern corn leaf blight
- maize streak - Mr. Kiriro stated that maize streak constitutes a serious threat to maize production in the Kakamega area.

Finger millet:
A blast disease of the panicle in which the causal agent is unknown.

Sorghum:
Most of the crop had been harvested. Ergot was of high incidence on the late maturing cutivars.

Aug 16 Visited the International Center for Insect Physiology and Ecology (ICIPE) at Mbita point. Dr. Wunnithan, Entomologist, and Mr. Okiri, Administrative Officer, hosted the tour. The principal objective of this center is to increase food production in developing countries in tropical regions. Maize, sorghum, cowpea and bananas are the major crops being researched with lesser emphasis on rice. Besides the main center in Mbita Point, ICIPE has scientists working on rice insects at IRRI, a sorghum improvement project in Somalia and a maize streak project in Lusaka, Zambia.

Dr. Vincent Okoth is the ICIPE scientist with responsibility for maize streak. It is my understanding that the ICIPE program in Zambia is screening maize lines for leafhopper tolerance and also to the maize streak virus.

ICIPE will soon begin a Ph.D. program in Entomology. It will be a three year program with six months of course work being offered at Mbita Point. They expect 8 to 10 students to be in the program and they have agreements with 14

African Universities including universities at Kenyatta, Moi, Makerere and Port Harcourt.

The M.S. program at ICIPE will be offered at the University of Zimbabwe, Harare.

Aug 17 Spent most of the morning at the ICIPE Doctor's office for treatment of a stomach ailment.

Discussed maize borer projects with Mats Svensson, Ph.D. student from Sweden and Dr. S. Firempong, ICIPE.

Aug 18 Departed Kisumu around 10:00 a.m. after exchanging money and arrived at Kitale about 1:30 p.m. Discussed research with Mr. Ochieng as rain cancelled the tour of his nurseries.

Aug 19 Surveyed maize fields in the vicinity of Kenya Seed company.

Aug 20 Rested at Kitale Club Hotel.

Aug 21 Visited with Mr. Ochor and Mr. John Kedera, plant pathologists, concerning plant pathology research projects at Kitale. Considerable time was spent examining laboratory equipment and future needs that will enable them to have sufficient equipment to conduct research.

Maize fields in the Eldoret area were surveyed for diseases. Limited diseases were observed with the incidence of maize streak being less than 5%. Lack of nitrogen fertizer appeared to be the major problem.

Aug 22 Dr. Bill Compton and myself toured the Kenya Seed Company plots. We were hosted by Mr. Michael Ngugi, Research Agronomist and Miss Pamela Rahedi, Research Officer. Considerable time was spent examining the maize inbreds and cultivars. At the request of Kenya Seed Company, I collected leaf samples from dry beans and will attempt to identify the causal agent. This disease has defoliated the leaves despite timely fungicide applications.

Kenya Seed Company has an excellent seed quality program as they provide seed that has good vigor and free as humanly possible of seed-borne diseases. They provide additional services to their farming clientele by recommending optimum cultural practices and chemical and pest management strategies.

Aug 23 Dr. Mary Compton, Mr. Nicodemus Mwania (Senior Maize Agronomist), Mr. Achieng (Agronomist) and myself visited on farm trials of two farmers near Moi's bridge. Except for maize streak, diseases were minimal and yields will be in the vicinity of 25-40 bags/hectare. Incidence of maize streak ranged from 8 to 20%.

In the afternoon, Mr. John Kedera, Dr. Mary Compton and myself drove to Kakamega. Along with Mr. Henry Wakhonya

(maize agronomist) and Mr. O. M. Odongo (maize breeder), we visited the striga research plots at Lubina village (west of Kakamega about 30 km). Maize streak was of high incidence with averages between 23 and 30%. More importantly, the plants were infected early in the season which resulted in stunted and barren plants. Maize streak incidence at the station in Kakamega was less than 1%.

Aug 24 Visited Mr. Tobias Ochor in the district hospital. Mr. Ochor was seriously affected with a respiratory ailment and his proposed Ph.D training in my laboratory will have to be postponed indefinitely.

Discussed future collaborative research projects with Drs. Bill and Mary Compton, Mr. Ochieng and Mr. Muthoka.

Departed Kitale at 11:30 a.m. for Kisumu. Left Kisumu at 5:30 via Kenya Airways for Nairobi (arrived 6:30 p.m.).

Aug 25 Scheduled an appointment with Dr. C. G. Ndiritu, Director, KARI to inform him of the results of my survey throughout the maize growing regions of Kenya for the past six weeks. He was informed of the maize streak problem that poses a serious constraint to maize production at the present and in the future. Dr. Ndiritu requested that I write a synopsis of recommendations and my trip itinerary.

A meeting was also held with Dr, John Lynam, Agricultural Sciences Representative, The Rockefeller Foundation. Dr. Lynam wanted an update on the maize streak situation in Kenya.

Aug 26 Worked on the synopsis to Director Ndiritu.

Aug 27 Rested at Hilton Hotel.

Aug 28 Worked at the MIAC office in Kenya House. The secretary began typing my reports.

Aug 29 Worked in the MIAC office.

Aug 30 A meeting was scheduled to consult with Dr. Brhane Gebrekidan, Regional Maize Breeder, CIMMYT, to discuss potential maize germplasm that is available to be used in Kenya maize production. At the present, it appears that only the CIMMYT germplasm developed in Zimbabwe and the IITA germplasm developed in the midaltitude area of Cameroun are the only viable materials.

Dr. Gebrekidan and I drove to Muguga to revisit the station to observe the maize germplasm. We were joined by Mr. Jackson Njuguna on a tour of the plots. Maize streak tolerant germplasm from Reunion and IITA are unacceptable as they are only adapted for low altitudes. Germplasm from Burundi possesses tolerance (not immunity) to maize streak but is not acceptable at the present since this is a

segregating population and it will take 4-5 years to select for desirable lines.

Aug 31 Met with Dr. Cecil McFarland, USAID, Nairobi; for a briefing on the current maize disease situation in Kenya including maize streak.

Met with Dr. Larry Empig to discuss available maize streak germplasm and screening materials for resistance.

Sept 1 Worked in the MIAC office.

Sept 2 Worked on compiling a list of scientific equipment needs for laboratories in Mtwapa, Katumani, Embu, Kitale and Kakamega.

Met with Dr. Larry Empig in the afternoon. He and Linus Muriithi has just harvested the maize plants of a local cultivar, Kikuyu, that seemingly had tolerance to maize streak.

Sept 3 Worked in hotel on trip report including the section on recommendations as suggested by Dr. Daugherty.

Sept 4 Finalized reports and made preparations to leave Kenya.